DiscoverX

User Manual

PathHunter®

Detection Kit

Use with PathHunter Cell Lines or Cell Lines Made with ProLabel® /ProLink™ Vectors

Catalog No. 93-0001 Series

Please read entire booklet before proceeding with the assay.
For additional information or Technical Support see contact information below.

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Overview

The PathHunter Detection Kit is to be used with PathHunter Cell Lines or cell lines made with ProLabel®/ProLink™ expression or cloning vectors. The kit is easy-to-use and has been successfully run in both 96-well and 384-well microplate format. The resulting signal is chemiluminescent and is read with any standard plate reader.

PathHunter products utilize the Enzyme Fragment Complementation (EFC) technology for studying protein-protein interactions, protein translocation, receptor trafficking, and receptor internalization, all of which involve the complementation of two β-galactosidase (β-gal) fragments. The larger β-gal fragment is called Enzyme Acceptor (EA) and the smaller fragment is termed Enzyme Donor (ED; also known as ProLabel or ProLink). The two fragments are inactive when apart. However, when they complement, they form a functional enzyme that hydrolyzes the substrate to generate a chemiluminescent signal. The detection reagents do not contain either the EA or ED fragments of β-gal, and therefore must be used in conjunction with cells that express both components in order to obtain a detectable signal.

Materials Provided and Storage Conditions

Catalog Number	93-0001S	93-0001M	93-0001	93-0001L	93-0001XL
Number of Plates	1 Plate	2 Plates	10 Plates	100 Plates	500 Plates
96-well, No. of data points	~100	~200	~1000	n/a	n/a
384-well, No. of data points	~400	~800	~4,000	~40,000	~200,000
Kit Components	Volume in each bottle (mL)				
Cell Assay Buffer	5.7	11.4	57	418	1825
Substrate Reagent 1	1.5	3	15	110	480
Substrate Reagent 2	0.3	0.6	3.0	22	96
Total Kit Volume	**7.5**	**15**	**75**	**550**	**2401**

Upon arrival, store reagents at -20°C. The detection kit is stable until the expiration date indicated on the kit box outer label. Thaw reagents at room temperature before use. After thawing, store reagents for up to 1 month at 2-8°C. For longer term storage, aliquots of all the components may be re-frozen once in opaque containers at -20°C.

Additional Materials Needed

Required Materials
PathHunter cell line expressing EA and ED
15 mL Polypropylene tube (or larger, if necessary)
Multimode or luminescence plate reader
Disposable reagent reservoir (Thermo Scientific, Cat. No. 8094 or similar)

Assay Detection Protocol

The following procedure details the assay detection protocol to be used with PathHunter cell lines. Please refer to your cell-line user manual for specific cell line and assay preparation details. Detection reagents must be prepared as a working solution prior to use. Once prepared, the working solution is stable for up to 24 hours at room temperature with no impact on assay performance.

Working Detection Solution for 96 or 384-well Format		
Components	Volume Ratio	Volume per Plate (mL)
Cell Assay Buffer	19	5.7
Substrate Reagent 1	5	1.5
Substrate Reagent 2	1	0.3
Total Volume		**7.5**

1. Prepare a stock of Working Detection Solution in a 15 mL polypropylene tube or reagent reservoir by mixing 19 parts of cell assay buffer, 5 parts of Substrate Reagent, and 1 part of Substrate Reagent 2.

Make stock within 24 hours of use.

2. Add 55 µL (for 96-well format) or 12.5 µL (for 384-well format) [equivalent to 50% of the assay volume] of Working Detection Solution to all wells of the assay plate.

Do not pipette up and down in the vial to mix or vortex plates.

3. Incubate assay plate for 1 hour at room temperature in the dark.

Working Detection Solution is light sensitive, thus incubation in the dark is necessary.

4. Read samples on a standard luminescence plate reader at 0.1 to 1 second/ well for photomultiplier tube readers or 5-10 seconds for imager. The actual signal characteristics over time are affected by lab conditions such as temperature. The user should establish an optimal read time accordingly. Luminescence readout collects signal from all wavelengths. Some instrument manufacturers may include a cutoff filter at high wavelengths, but usually no wavelength setting is needed for luminescence readout.

After addition of reagents, the samples should be read within 2 hours.

5. Data analysis can be performed using your choice of statistical analysis software (e.g. GraphPad Prism, Molecular Devices Softmax Pro, Biotek Instruments Gen5, Microsoft Excel, etc.).

For additional information or technical support, see contact information below.

Supplemental Information

Instrument Compatibility Chart	
Compatible with any luminometer. Select examples indicated below.	
Berthold Technologies: Mithras LB940, CentroLIApc	**Molecular Devices:** FLIPR, SpectraMax M3/ M4/M5/M5e, FlexStation 3, SpectraMax L
Biotek: Synergy 2	**Perkin Elmer:** TopCount, Victor 2 or V, Fusion, LumiCount, Envision, Micro-beta (Trilux), Viewlux, Northstar, EnSpire
BMG: PheraStar, Cytostar, LumiStar	**Promega:** GloMax systems
Caliper: LabChip 3000 & EZ Reader	**Tecan:** Ultra, Evolution
GE: LEAD seeker, Farcyte	**Thermo Scientific:** Luminoskan Ascent
Hamamatsu: FDS6000, FDSS/RayCatcher	**Turner BioSystems:** Modulus Microplate

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